Collection

Generic Collections :-

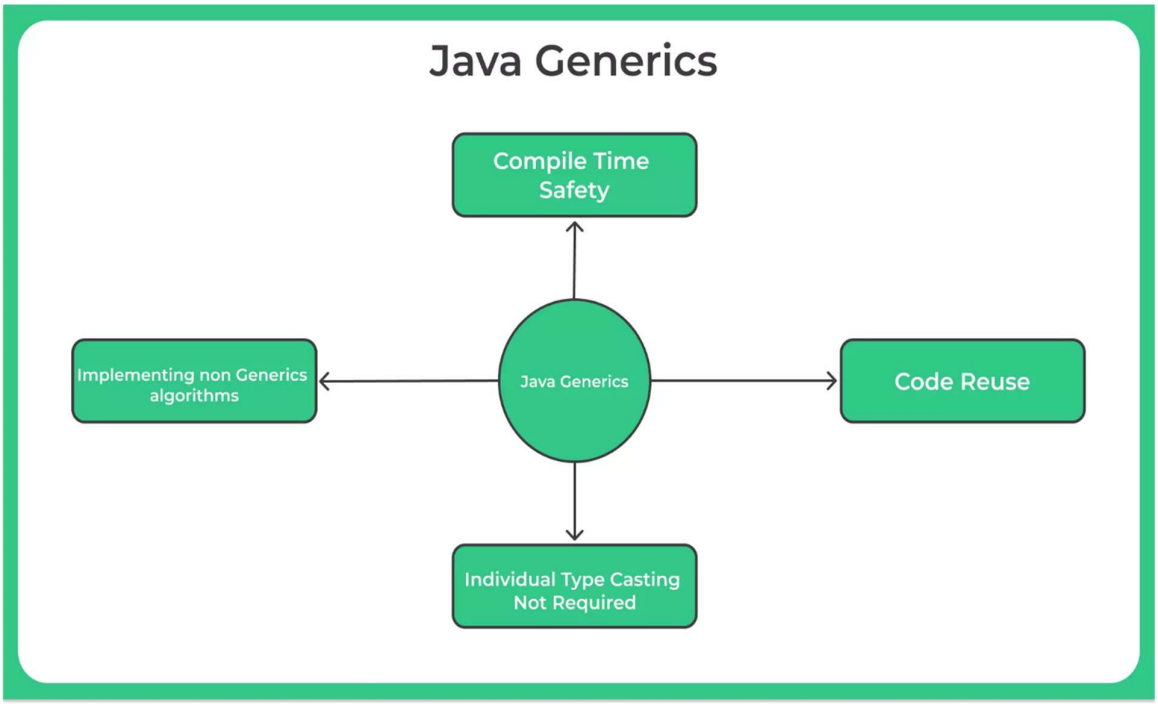
- A Generic Collection in Java allows you to specify the **type of objects** that a collection can hold.
- There are store in **same type of data.**
- It can using **Homogeneous** Object.

Syntax of Generic Collection :-

CollectionType<DataType> collectionName = **new** CollectionType<>();

OR

ArrayList<String> name = **new** ArrayList<String>();



Implementation of Generic Collection :-

Input :-

```java
package Collection_Framework;

import java.util.ArrayList;
import java.util.Iterator;

public class Generic_Collection {

    public static void main(String[] args) throws InterruptedException {

        ArrayList<String> name = new ArrayList<String>();
        name.add("Akshay");
        name.add("Nikhil");
        name.add("Akash");
        System.out.println("Generic Collection is : "+name);

        // Sorting
        name.sort(null);
        System.out.println("\nGeneric Collection after sorting is :"+name);


        // Iteration
        Iterator it = name.iterator();
        System.out.println("\nSorting of Collection is : ");
        while(it.hasNext())
        {
            System.out.println(it.next());
            Thread.sleep(1000);
        }
    }
}
```

Output :-

```
Generic Collection is : [Akshay, Nikhil, Akash]

Generic Collection after sorting is :[Akash, Akshay, Nikhil]

Sorting of Collection is :
Akash
Akshay
Nikhil
```

Linked List :-

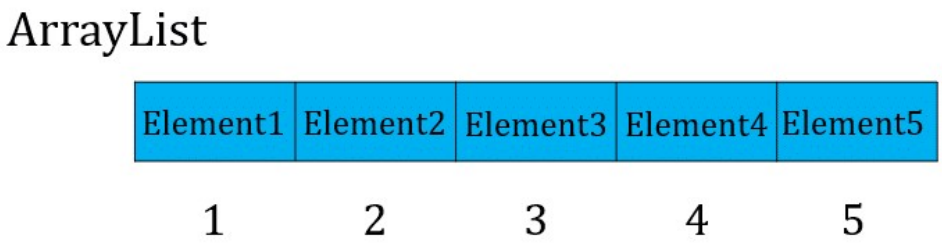
- LinkedList in Java is a class in the **java.util** package that implements both the **List** and **Deque** interfaces.
- It is a **doubly linked list**, meaning each element (node) has links to both the previous and next nodes.

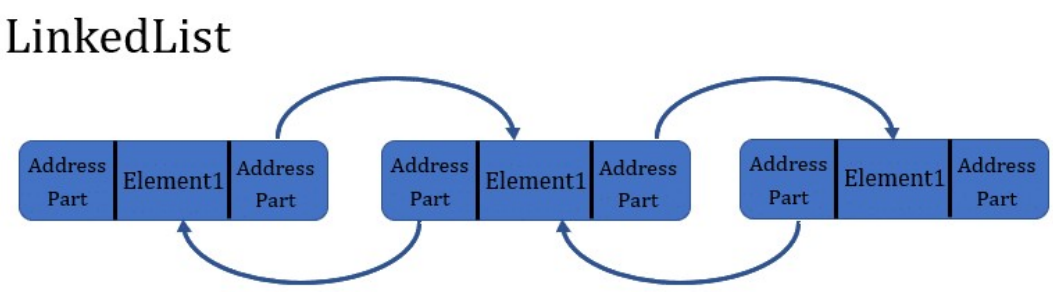
Syntax of LinkedList :-

```
LinkedList<Type> list = new LinkedList<Type>();
```

- LinkedList is ideal when your program involves **frequent insertion and deletion** of elements.
- It **maintains order**, allows **duplicates**, and provides **queue/deque functionalities**.

Difference between ArrayList and LinkedList :-





Implementation of LinkedList :-

Input :-

```java
package Collection_Framework;

import java.util.Iterator;
import java.util.LinkedList;

public class linkedlist {

    public static void main(String[] args) {

        LinkedList l = new LinkedList();
        l.add("nikhil@77gmail.com");
        l.add(567);
        l.add("Nikhil");
        l.add("#");

        System.out.println(l.offer(76));
        l.add(81.2);
        System.out.println("Peek : "+l.peek());
        System.out.println("Poll : "+l.poll());
        System.out.println("LinkedList : "+l);
    }
}
```

Output :-

```
true
Peek : nikhil@77gmail.com
LinkedList : [nikhil@77gmail.com, 567, Nikhil, #, 76, 81.2]
```

Vector :-

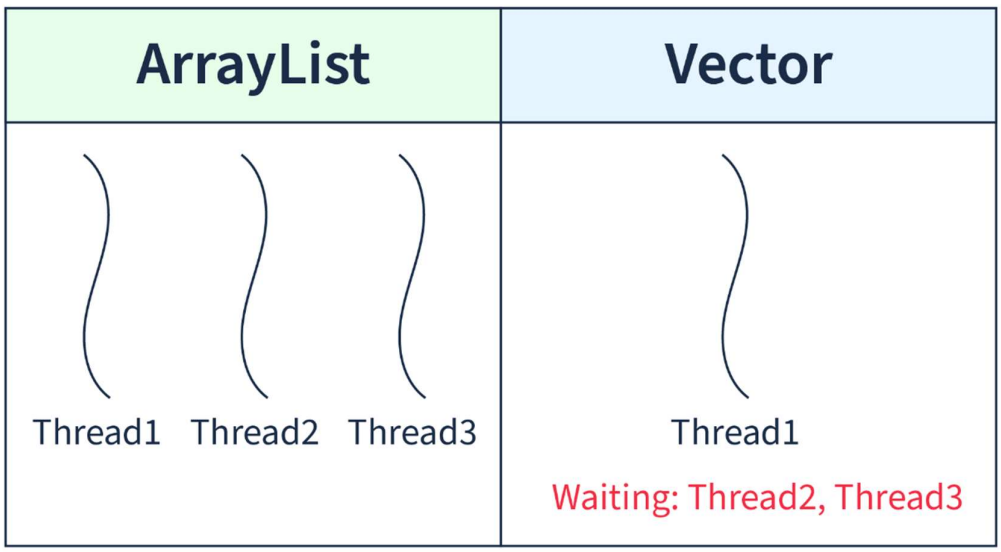
- Vector is a **legacy class** in Java that implements a **dynamic array** similar to ArrayList, but **synchronized.**
- It is part of the **java.util** package and implements the **List** interface.

Syntax of Vector :-

```
Vector<Type> vector = new Vector<Type>();
```

- Vector is a **thread-safe**, **dynamic array**.
- It's rarely used today — replaced by **ArrayList** and **Collections.synchronizedList()** for modern Java code.
- Still useful when **synchronization** is required by default.

Difference between ArrayList and Vector :-



Implementation of Vector :-

Input :-

```java
package Collection_Framework;

import java.util.Iterator;
import java.util.Vector;

public class Vector_demo {

    public static void main(String[] args) {

        Vector<Integer> rollno = new Vector<Integer>();
        rollno.add(90);
        rollno.add(56);
        rollno.add(80);

        // Iterator the object of collection change the value of index 1 to 60

        rollno.set(1, 60);

        for(Integer i:rollno)
        {
            System.out.println(i);
        }
    }
}
```

Output :-

```
90
60
80
```

Task 1

Input :-

```java
package Collection_Framework;

import java.util.ArrayList;

public class task_1 {

    public static void main(String[] args) {

        ArrayList al = new ArrayList();
        al.add(45);
        al.add("Nikhil");
        int flag=0;

        for(int i=0;i<al.size();i++)
        {
            if(al.get(i).equals("Nikhil"))
            {
                flag++;
            }
        }
        if(flag==0)
        {
            System.out.println("Nikhil object does not exist....!");
        }
        else
        {
            System.out.println("Nikhil object exist....!");
        }
    }
}
```

Output :-

```
Nikhil object exist....!
```

Task 2

Input :-

```java
package Collection_Framework;

import java.util.ArrayList;

public class task_2 {

    public static void main(String[] args) {

        ArrayList al = new ArrayList();
        al.add(45);
        al.add("y");
        al.add("Pune");
        al.add(45.6);
        al.add(98747290474635383l);

        for(int i=0; i<al.size();i++)
        {
            if(i==3)
            {
                al.add(i, 78923.23);
            }
        }
        System.out.println("ArrayList is : "+al);
    }
}
```

Output :-

```
ArrayList is : [45, y, Pune, 78923.23, 45.6, 98747290474635383]
```